MICHAEL FRYER

Data Scientist & Software Engineer | M.Sc. Applied Data Science Candidate

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Residence Permit in Germany | English (Native) | German (A2)
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Software engineer transitioning into data science, currently pursuing an M.Sc. in Applied Data Science at Frankfurt School of Finance & Management. Experienced in Python-based machine learning, statistical analysis, and data visualization, with a professional background in full-stack software development. Seeking to apply analytical and engineering skills to data-driven roles in industry.

EDUCATION

M.Sc. in Applied Data Science
September 2025—Present
Frankfurt School of Finance & Management
Frankfurt am Main, Germany
Expected graduation: Summer 2027 | Relevant coursework: Machine Learning, Statistical Modeling, Data Engineering, Deep Learning

B.Sc. in Software Engineering and B.Sc. in Computer Science
August 2015—May 2019
Montana Tech
Butte, MT, USA

EXPERIENCE

Machine Learning Teaching Assistant
January 2026—Present
Frankfurt School of Finance & Management
Frankfurt am Main, Germany

- Designed supplementary exercises for a bachelor-level Machine Learning course covering supervised learning, neural networks, and model evaluation.
- Supported students with Python-based ML implementations, debugging model pipelines and explaining statistical concepts during weekly office hours.

Associate Game Programmer
October 2019—July 2021
Heavy Iron Studios
Manhattan Beach, CA, USA

- Worked closely with artists to transform their UI mock-ups into a functional user experience.
- Extended Unreal Engine's replication graph to meet our network performance requirements.
- Shipped production-ready features across multiple Unreal Engine 4 patch releases in preparation for a full commercial game launch.
- Built and deployed the Live Operations portal using Django and AWS, enabling real-time content management for a live-service title post-launch.

Software Engineer
September 2018—August 2019
Butte Food Bank
Butte, MT, USA

- Designed and developed FRED, a full-stack inventory and reporting system that streamlined operations for the Butte Food Bank.
- Developed the front-end application of FRED using WPF Forms.
- Expanded reporting features to help the Food Bank meet government regulations.

PROJECTS

Urban Mobility Risk Analysis *| Python, Pandas, DBSCAN, Contextily*
Applied DBSCAN clustering to a geospatial traffic accident dataset for Frankfurt am Main to identify and rank high-risk zones. Visualized results as choropleth maps using Contextily and Pandas for exploratory data analysis and stakeholder reporting.

Arcadia Website *| NextJS, Vercel, Neon*
A NextJS application that leverages Google's YouTube API to aggregate video uploads from members of Arcadia and post to social media.

TECHNICAL SKILLS

Machine Learning | Deep Learning | Data Analysis | Python | PyTorch | TensorFlow | Scikit-learn | Pandas | NumPy | SciPy | SQL | Altair | Jupyter | AWS | Docker | Linux/Bash | Git | TypeScript | C/C++

Analytical Thinking | Problem Solving | Attention to Detail | Communication | Collaboration